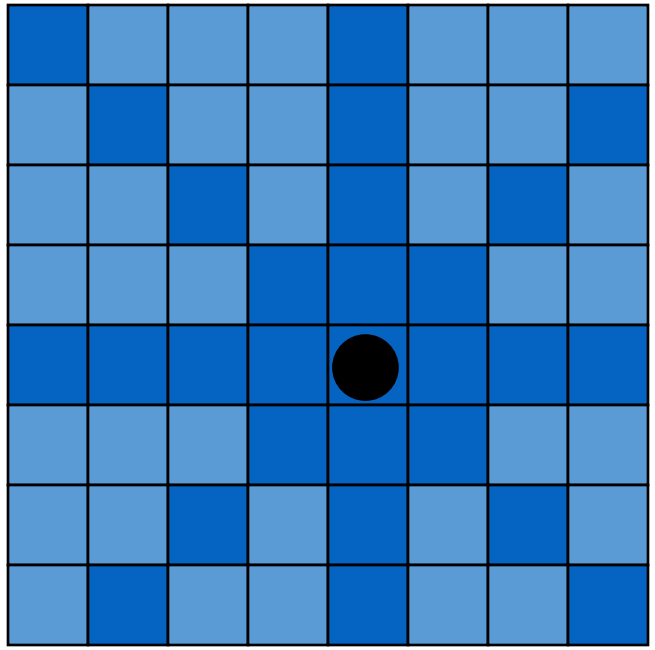
Genetic Algorithm

Previous Lecture

- Components pf Genetic Algorithm
  - Encoding principles (gene, chromosome)
  - Initialization procedure (creation)
  - Selection of parents (reproduction)
  - Genetic operators (mutation, recombination)
  - Evaluation function
  - Termination condition

n-Queen Problem

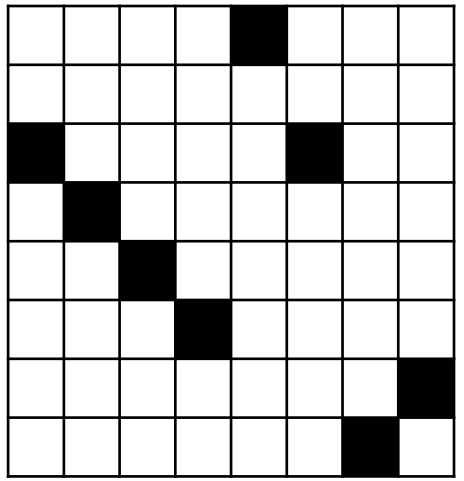
Example: the 8 queens problem



Place 8 queens on an 8x8 chessboard in such a way that they cannot check each other

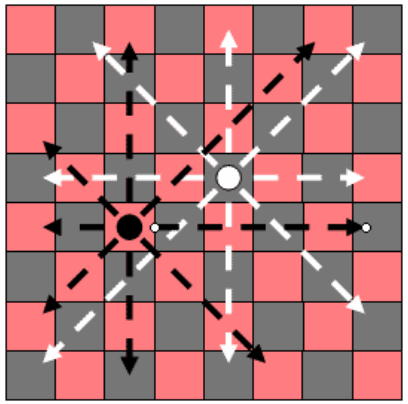
8-Queens Problem

- State = position of 8 queens each in a column
  - 3 4 5 6 1 3 8 7



- Start with *k* randomly generated states (population)
- Evaluation function (fitness function).
- Higher values for better states.

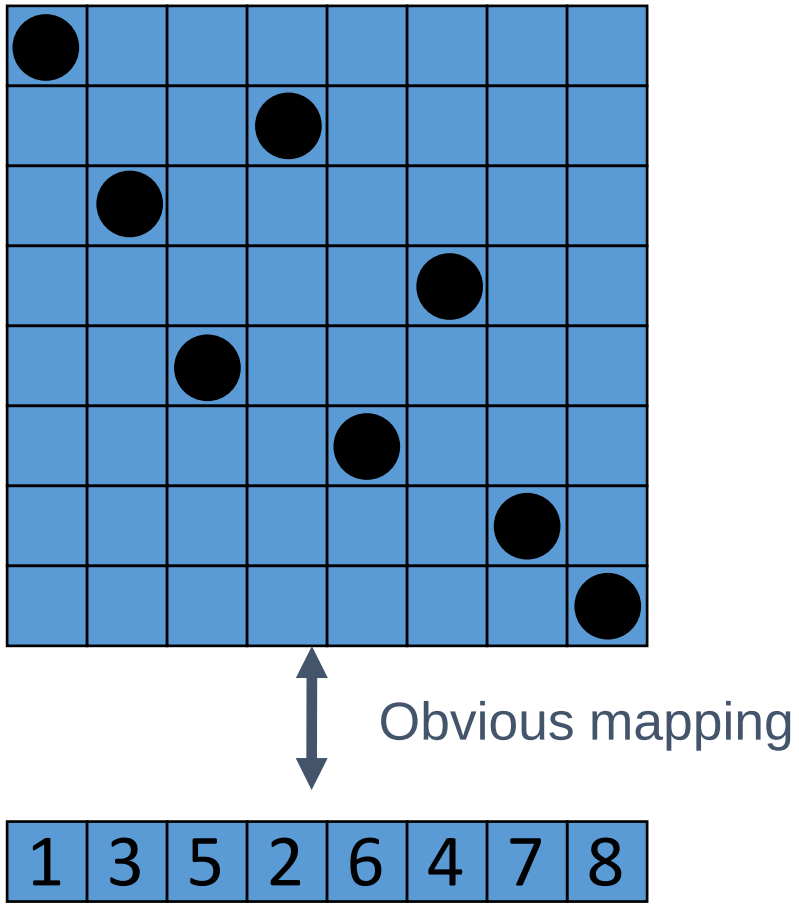
- Produce the next generation of states by “simulated evolution”
- Random selection
  - Crossover
  - Random mutation



The 8 queens problem: representation

Phenotype:
a board configuration

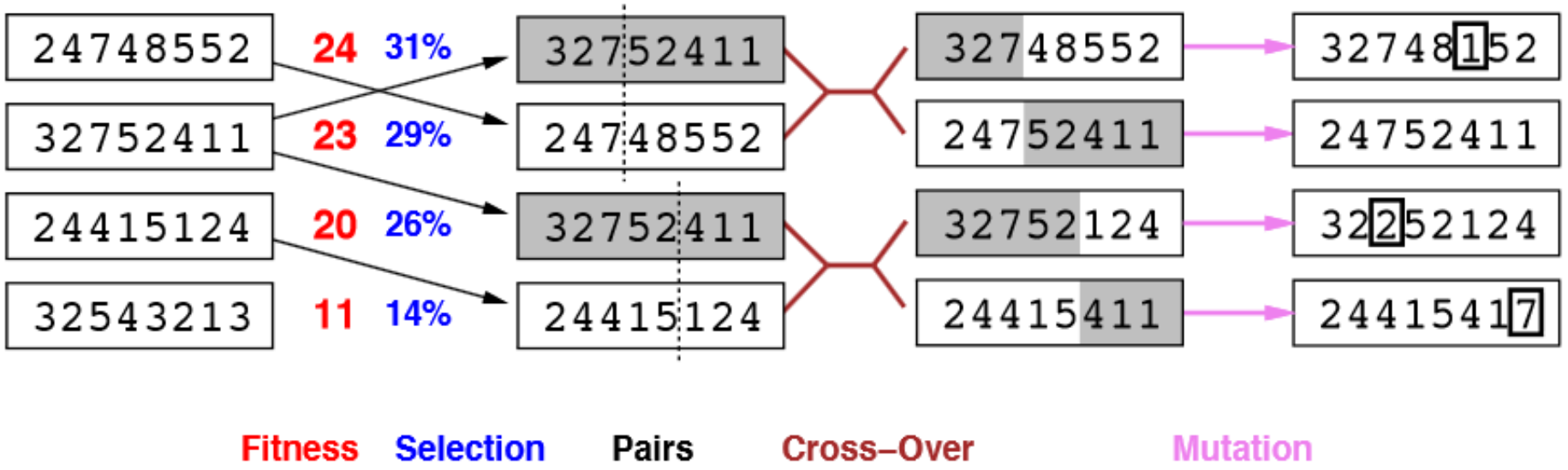
Genotype:
a permutation of
the numbers 1 - 8



8 Queens Problem: Fitness evaluation

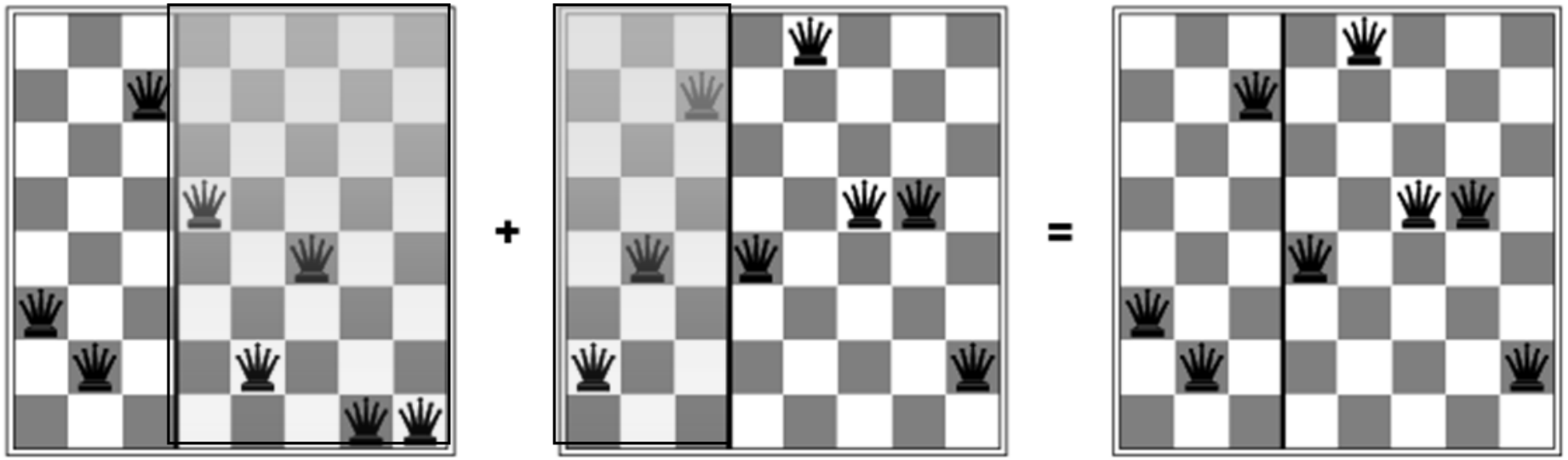
- Penalty of one queen:
  - the number of queens she can check.
- Penalty of a configuration:
  - the sum of the penalties of all queens.
- Note: penalty is to be minimized
- Fitness of a configuration:
  - inverse penalty to be maximized

8-Queens Problem



Note: $24 / (24+23+20+11) = 31\%$

Effect of Crossover on 8-Queens



32752411 + 247**48552** = 32748552

Has the effect of "jumping" to a completely different new part of the search space (quite non-local)

The 8 queens problem: Selection

- Parent selection:
  - Pick 5 parents and take best two to undergo crossover
- Survivor selection (replacement)
  - When inserting a new child into the population, choose an existing member to replace by:
  - sorting the whole population by decreasing fitness
  - enumerating this list from high to low
  - replacing the first with a fitness lower than the given child

Travelling Salesman Problem (TSP)

Traveling Salesman Problem (TSP)

[0, 4, 1, 2, 6, 5, 7, 3] =
["Brighton", "Liverpool", "Bristol", "Cambridge", "Manchester", "London", "Oxford", "Glasgow"]

- The exact application involved finding the shortest distance to fly between eight cities without visiting a city more than once.
- The table below shows the distances between each city in kilometers.

	Brighton	Bristol	Cambridge	Glasgow	Liverpool	London	Manchester	Oxford
Brighton	0	172	145	607	329	72	312	120
Bristol	172	0	192	494	209	158	216	92
Cambridge	145	192	0	490	237	75	205	100
Glasgow	607	494	490	0	286	545	296	489
Liverpool	329	209	237	286	0	421	49	208
London	72	158	75	545	421	0	249	75
Manchester	312	216	205	296	49	249	9	194
Oxford	120	92	100	489	208	75	194	0

TSP – Initialization

- **Initialization**
  - Each individual featuring an integer representation of a route between the eight cities with no repetition.
  - A corresponding array with the string equivalent of these indexes is created to output when a solution is found.

[0, 4, 1, 2, 6, 5, 7, 3] =
["Brighton", "Liverpool", "Bristol", "Cambridge", "Manchester",
"London", "Oxford", "Glasgow"]

- **Fitness Function**
  - A fitness function calculates the total distance between each city in the chromosome’s permutation.

- Consider the *Traveling Salesman Problem* (*TSP*) in which a salesman aims to visit *n* cities *exactly once* covering the least distance
- Starting at any given node, choose from *n*–1 remaining nodes, then choose from *n*–2 remaining nodes, etc. Testing every possible route takes (*n*–1)**!** Steps
- Use a genetic algorithm to *evolve* a near-optimal solution to the TSP
- Label cities *A*, *B*, *C*, *D*, *E*, *F*, etc.
- Example circuits: *ABCDEF*, *BDAFCE*, *FBECAD*
- *How do we perform crossover operations?*
- Basic crossovers might result in invalid members of the population
  - e.g. combining *ABCDEF* and *BDAFCE* may result in *ABCFCE*

Another Example: Traveling Salesman Problem (TSP)

- The traveling salesman must visit every city in his territory exactly once and then return to the starting point; given the cost of travel between all cities, how should he plan his itinerary for minimum total cost of the entire tour?

- TSP $\in$ NP-Complete

- Note: we shall discuss a single possible approach to approximate the TSP by GAs

TSP (Representation, Evaluation, Initialization and Selection)

- A vector $v = (i_1\, i_2 \ldots\, i_n)$ represents a tour (v is a permutation of {1,2,…,n})
- Fitness f of a solution is the inverse cost of the corresponding tour
- Initialization: use either some heuristics, or a random sample of permutations of {1,2,…,n}
- We shall use the fitness proportionate selection

TSP (Crossover)

- Builds offspring by choosing a sub-sequence of a tour from one parent and preserving the relative order of cities from the other parent and feasibility
- Example:
- p_1 = (1 2 3 4 5 6 7 8 9) and
- p_2 = (4 5 2 1 8 7 6 9 3)
- First, the segments between cut points are copied into offspring
- o_1 = (x x x 4 5 6 7 x x) and
- o_2 = (x x x 1 8 7 6 x x)

TSP (Crossover-continued)

- Next, starting from the second cut point of one parent, the cities from the other parent are copied in the same order
- The sequence of the cities in the second parent is

9 – 3 – 4 – 5 – 2 – 1 – 8 – 7 – 6

- After removal of cities from the first offspring we get

9 – 3 – 2 – 1 – 8

- This sequence is placed in the first offspring
- o_1 = (2 1 8 4 5 6 7 9 3), and similarly in the second
- o_2 = (3 4 5 1 8 7 6 9 2)

TSP (Inversion)

- The sub-string between two randomly selected points in the path is reversed

Reverse Mutation

- Example:

(1 2|3 4 5 6 7|8 9) is changed into (1 2 7 6 5 4 3 8 9)

- Such simple inversion guarantees that the resulting offspring is a legal tour

The Traveling Salesman Problem

- Find a tour of a given set of cities so that
  - each city is visited only once
  - the total distance traveled is minimized

Representation

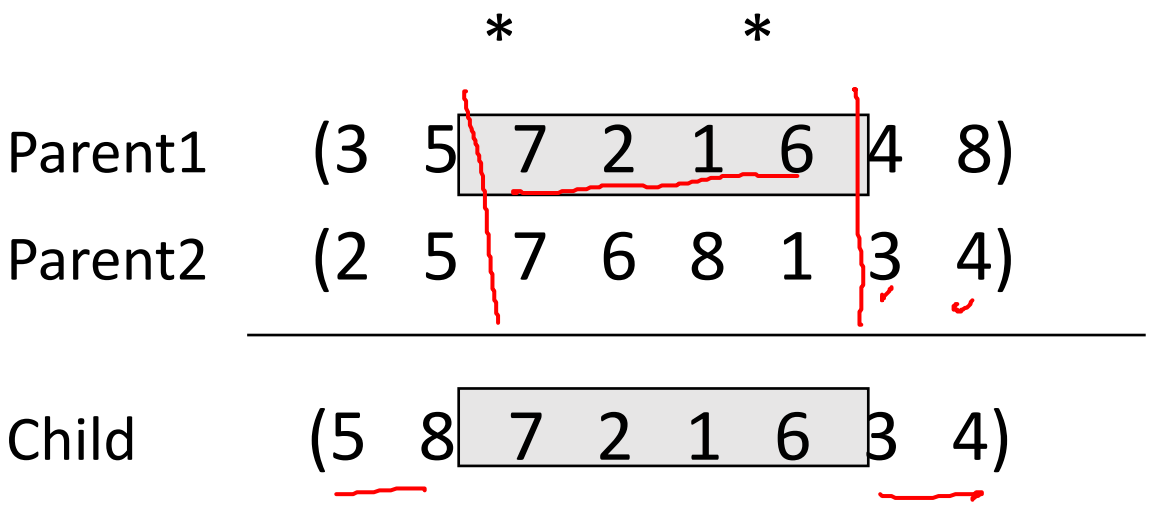
- Representation is an ordered list of city numbers

1) London 3) Dunedin 5) Beijing 7) Tokyo
2) Venice 4) Singapore 6) Phoenix 8) Victoria

City List 1 (3 5 7 2 1 6 4 8)
City List 2 (2 5 7 6 8 1 3 4)

Crossover

- Crossover combines inversion and recombination:



- This operator is called order-based crossover.

TSP (Crossover1)

- Builds offspring by choosing a sub-sequence of a tour from one parent and preserving the relative order of cities from the other parent and feasibility
- **Example**:
  - p_1 = (1 2 3 4 5 6 7 8 9) and
  - p_2 = (4 5 2 1 8 7 6 9 3)
- First, the segments between cut points are copied into offspring
  - o_1 = (x x x 4 5 6 7 x x) and
  - o_2 = (x x x 1 8 7 6 x x)

TSP (Crossover2)

- Next, starting from the second cut point of one parent, the cities from the other parent are copied in the same order
- The sequence of the cities in the second parent is

  9 – 3 – 4 – 5 – 2 – 1 – 8 – 7 – 6

- After removal of cities from the first offspring we get

  9 – 3 – 2 – 1 – 8

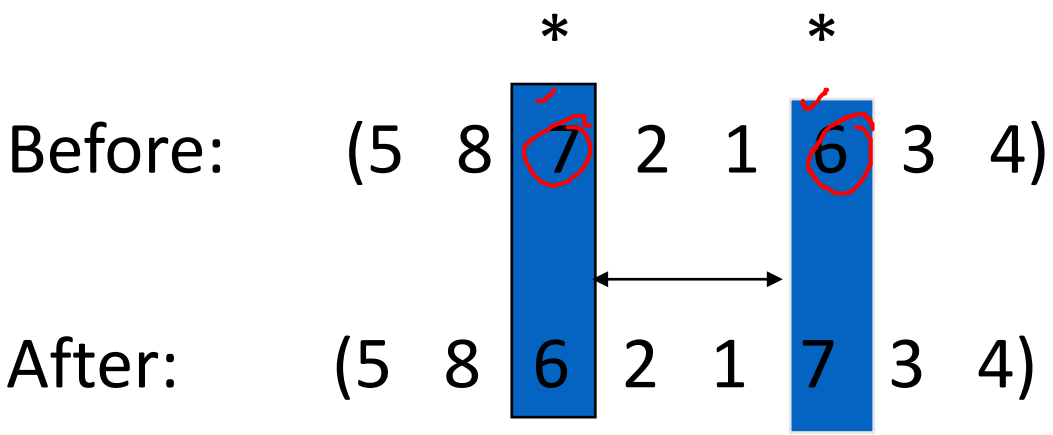
- This sequence is placed in the first offspring
  - o_1 = (2 1 8 4 5 6 7 9 3), and similarly in the second
  - o_2 = (3 4 5 1 8 7 6 9 2)

TSP (Inversion)

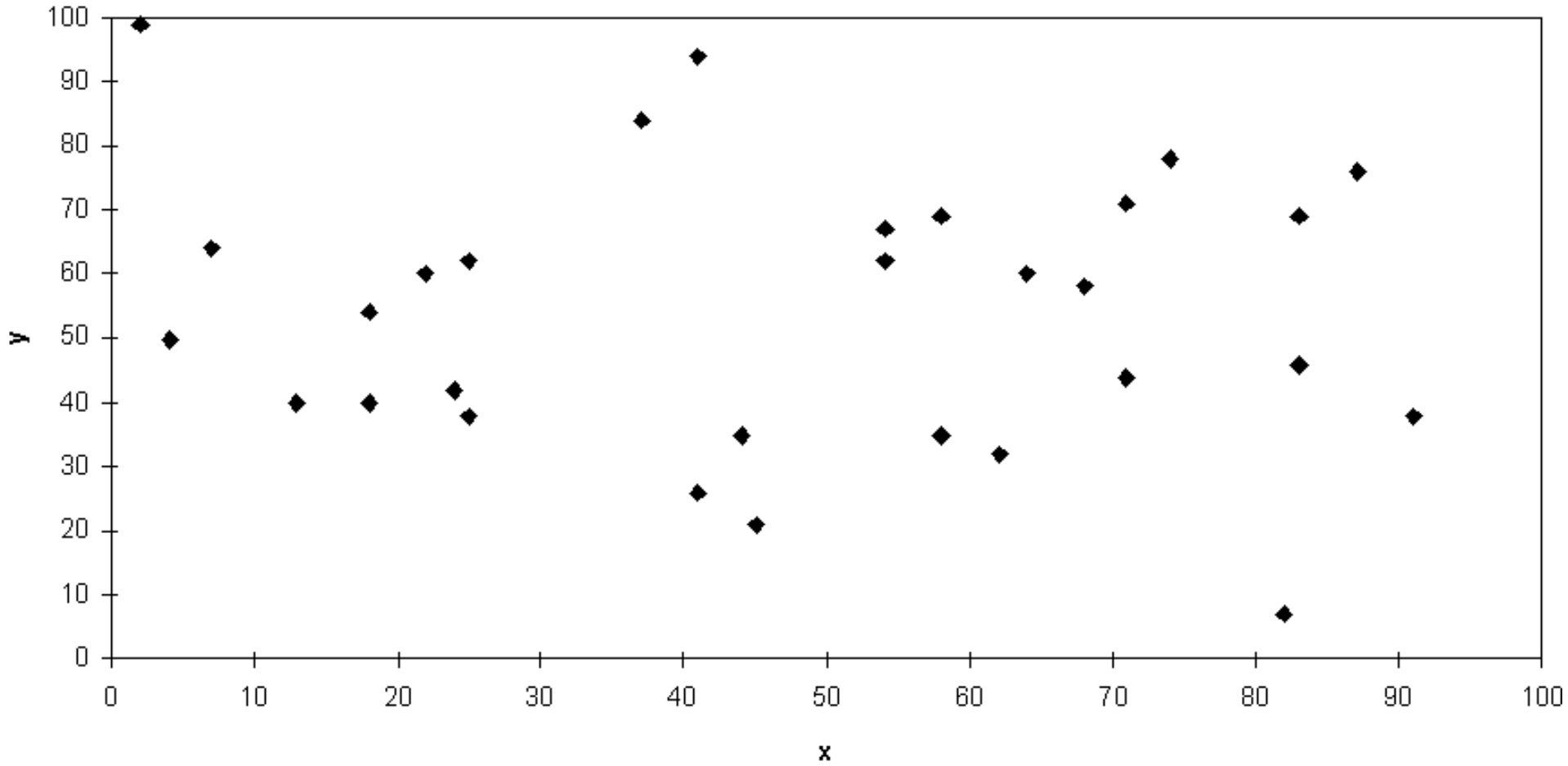
- The sub-string between two randomly selected points in the path is reversed
- Example:
  - (1 2 3 4 5 6 7 8 9) is changed into (1 2 7 6 5 4 3 8 9)
- Such simple inversion guarantees that the resulting offspring is a legal tour

Mutation

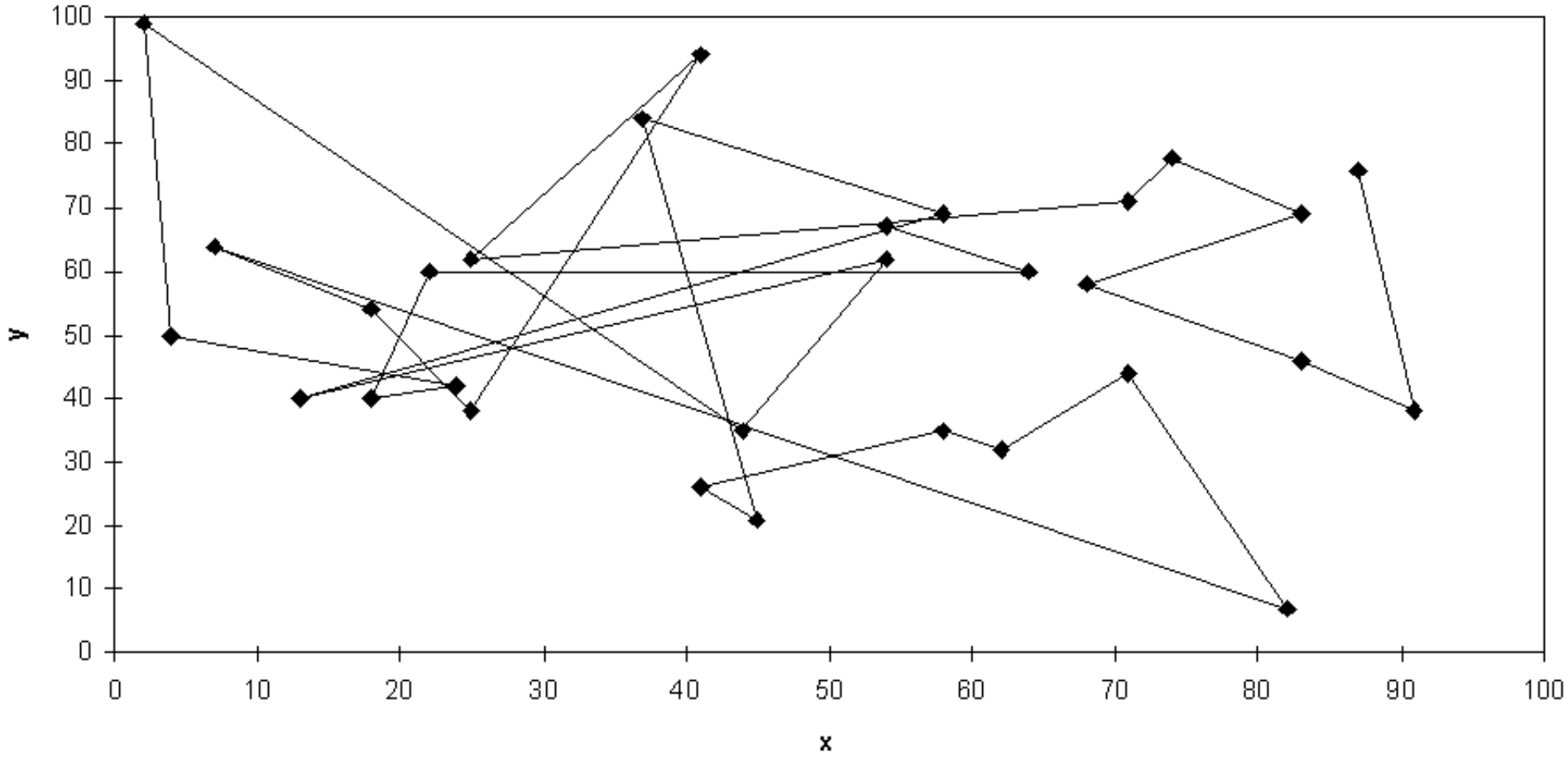
- Mutation involves reordering of the list:



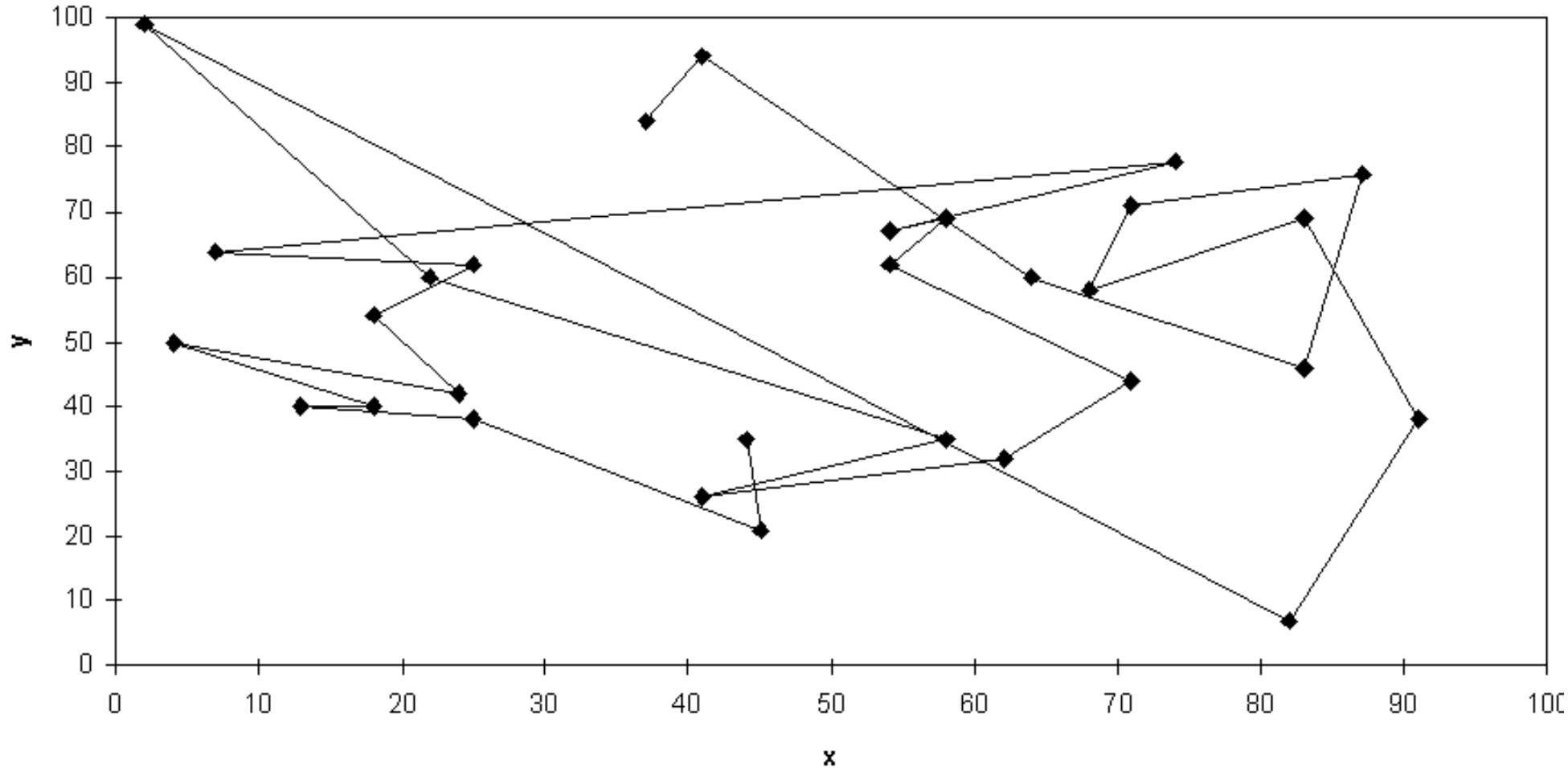
TSP Example: 30 Cities



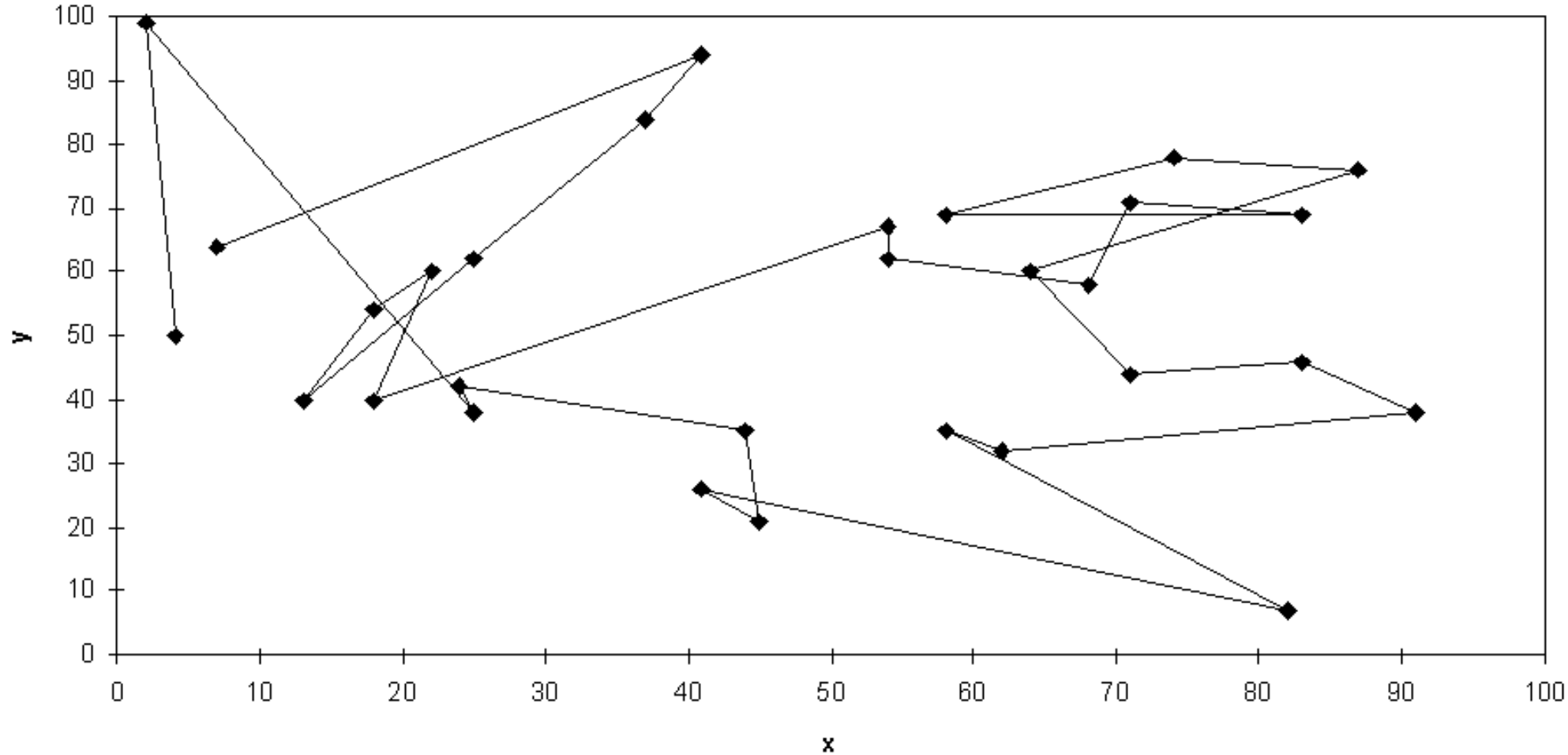
Solution $_i$ (Distance = 941)



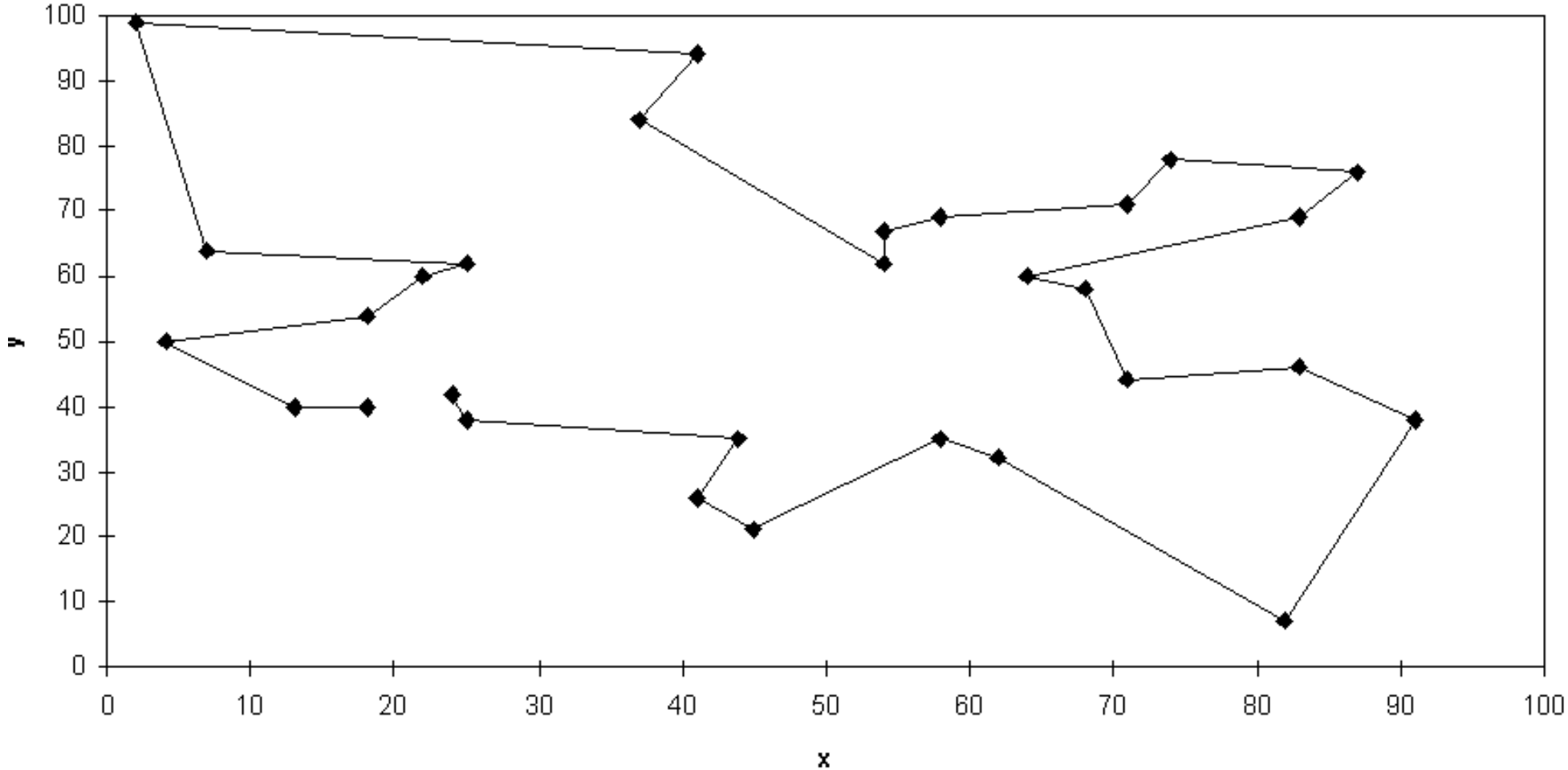
Solution $_j$ (Distance = 800)



Solution $_k$ (Distance = 652)



Best Solution (Distance = 420)



Practice Question – TSP

- Consider the problem of finding the shortest route through cities, such that each city is visited only once and in the end return to the starting city. Suppose we apply GA, in which genes represent links between pairs of cities.

  - How many genes will be used in a chromosome of each individual if the number of cities is 10?

  - For TSP problem, consider the following two routes as parents. Apply order based crossover on index 4 to 7 (index starts from 0).

  - The mutation for TSP can be defined as a swap between two indices. Now apply mutation at index 5 and 6 on both parents.